MATHEMATICS

University *of* NORTH ALABAMA

Spring 2026 Math Problem of the Week: Problem #14

Due noon Monday, April 27
Open to all UNA undergraduates

Directions: Deliver your written solution to me (Dr. Bryant) in CPTMTH 328 by noon Monday, April 27 or, if you cannot find me, you may leave it with Ms. Phifer in CPTMTH 377. All work must be your own. You are required to justify your response fully. If you utilize a well-known theorem, state the theorem's full name and proposition, showing that the conditions of the theorem are met. Good luck!

Problem #14: Two ants begin at distinct vertices of the complete graph K_n. At each time $t = 1, 2, 3, \ldots$, both ants simultaneously traverse a uniformly random edge incident to their current vertex, each choosing among the $n - 1$ available edges with probability $\dfrac{1}{n-1}$. They continue this process until time T, when both ants occupy the same vertex for the first time. Find the expected value of T.